

LESSON 6.8

SOLVING REAL-WORLD PROBLEMS INVOLVING PROPORTIONS



LESSON INTRODUCTION

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

LEARNING TARGETS

- I can solve problems using proportions.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can ratios and rates be used to relate and compare numbers in context?
- What kinds of relationships can proportions represent?
- How do you set up a proportion to represent a situation?

LESSON NARRATIVE

As the third lesson in the section on solving problems with proportions, this lesson builds off of the prior two, introducing more complicated contexts and creating and solving part-to-part and part-to-whole comparisons with proportions. The lesson begins with an exploration of a part-to-part comparison using fractions as parts before moving to problems that have more context so that students can plan what is important to structure a proportion to solve.

COMPONENT SUMMARY

6.8.1 WARM-UP (~5 MIN.)

Filling in the missing digits to create equivalent ratios

6.8.3 BRING IT TOGETHER (10 MIN.)

Comparing and contrasting part-to-part and part-to-whole proportions

6.8.5 WRAP-UP (~5 MIN.)

Proportion application in real-world context

6.8.2 EXPLORATION (15-20 MIN.)

Exploring the approach of solving with proportions

6.8.4 YOUR TURN! (10 MIN.)

Practice creating part-to-part and part-to-whole proportions to solve problems

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Calculators
- Fraction/decimal conversion table

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with differentiating between part-to-part vs part-to-whole comparisons in context.
- Students may use direct division instead of proportional division. For example, if it takes 2 people 4 hours to do a certain task, students may mistakenly think that it would take 1 person 2 hours rather than 8 hours.
- Students may use addition or subtraction instead of multiplication or division to find the scale factor.

THINGS TO CONSIDER

- While still using proportional reasoning, we are no longer converting to different units. Making this clear for students who are looking for conversion rates means discussions are now focused on finding a missing piece of a ratio to complete the proportion so that the incomplete ratio has the same value of the given ratio.
- In activities 6.8.2 Exploration, 6.8.4 Your Turn!, and 6.8.5 Wrap-Up, the scale factors are not whole numbers, which may hold some students up.

LITERACY AND LANGUAGE ROUTINES

Three Reads

Students can practice Three Reads in 6.8.2 Exploration before applying it during 6.8.4 Your Turn!. Reading the problem and question multiple times with a specific purpose for reading supports students' comprehension of the problem context, creates a plan for solving, and encourages meta-awareness of their use of math language.

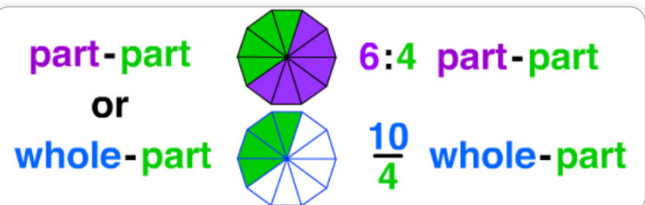
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 6.8.2 Exploration and 6.8.3 Bring It Together, students learn they need to read closely for relevant details of what each value in the problem means to uncover in the relationship between the values in order to create a proportion. The questions included help them see these relationships explicitly and flexibly in order to support both procedural and conceptual understanding.

DIFFERENTIATE INSTRUCTION

Consider visual scaffolding for comparing and contrasting part-to-part and part-to-whole relationships. Students will often be proficient in solving and calculating when given a proportion but struggle to find an entry point when interpreting a word problem. Consider using colors, household objects, anchor charts, and other strategies for creating as many entry points as possible for students to get started when problem-solving.



6.8.1 WARM-UP: PROPORTIONS WITH FRACTIONS

DIFFERENTIATE INSTRUCTION

During this Warm-Up, have students describe their proportion answers verbally, using a structure like the following:

“ $\frac{1}{2}$ pizza is to $\frac{1}{4}$ pizza like $\frac{1}{3}$ pizza is to $\frac{1}{6}$ pizza.”

Work in more precise language like “equivalent” to replace words such as “like.”



6.8.1 Warm-Up: Proportions with Fractions

- Place a digit in each blue box to make the equality statement true. Use only digits 1 to 9, at most one time each.

$$\frac{\overline{\overline{1}}}{\overline{\overline{\boxed{2}}}} = \frac{\overline{\overline{1}}}{\overline{\overline{\boxed{3}}}}$$

$$\frac{\overline{\overline{1}}}{\overline{\overline{\boxed{4}}}} = \frac{\overline{\overline{1}}}{\overline{\overline{\boxed{6}}}}$$

Other possible answers include:

$$\frac{\overline{\overline{1}}}{\overline{\overline{\boxed{2}}}} = \frac{\overline{\overline{1}}}{\overline{\overline{\boxed{4}}}}$$

$$\frac{\overline{\overline{1}}}{\overline{\overline{\boxed{6}}}} = \frac{\overline{\overline{1}}}{\overline{\overline{\boxed{3}}}}$$



6.8.2 Exploration: Which Information Should I Use?

Elena is making large batches of fruit punch and marinated chicken for her upcoming family reunion.

She makes fruit punch by mixing 5 parts cranberry juice with 3 parts apple juice.

1. Complete the statement to explain what this means by writing numbers in each blank.

Regardless of the unit she uses to measure, Elena will use a ratio of cranberry to apple juice that is 5 to 3. For example, if she is measuring in cups, she would add 3 cups of apple juice for every 5 cups of cranberry juice she adds.

2. Elena plans to make 40 cups of fruit punch. Work with your partner to complete the proportion that could be used to find how much apple juice she needs for the fruit punch, a .

$$\frac{3}{8} = \frac{a}{40}$$

3. What does the numerator of each ratio represent in the proportion?
The amount of the apple juice
4. What does the denominator of each ratio represent in the proportion?
The total fruit punch
5. How did you use the values 5 and 3 to determine the numerator and denominator of the first ratio?
3 is apple juice only
Total points are $3 + 5 = 8$ parts

6.8.2 EXPLORATION: WHICH INFORMATION SHOULD I USE?

DIFFERENTIATE INSTRUCTION

Three Reads

Consider a language routine like Three Reads to provide students scaffolding for making sense through the Exploration.

- First Read: Have students read through the Exploration in full without answering any questions.
- Second Read: Have students read through again, specifically circling or making note of questions that they are unsure how to unpack.
- Turn and Talk: Have students then compare with a partner or small group to clarify each other's questions and complete the activity.
- Third Read: Have students conduct a final read to reflect on what they have learned, ask questions, and correct any final errors.

ENRICHMENT

Ask students what the 40 represents in the proportion and what it is being compared to. This positions students to see that this question is asking them to find the total cups of juice and not just three cups of apple juice or five cups of cranberry juice.

6.8.2 EXPLORATION: WHICH INFORMATION SHOULD I USE? (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider bringing in measuring cups to provide students with visual cues or manipulatives for an additional entry point in making sense of setting up proportions.

ENRICHMENT

Notice question 8 provides the connection between scaling up and multiplication. Consider prompting students to explain why scaling up means multiplication and what that means for proportional relationships.

- What is the relationship between the scale factor and the proportion?
- Would you always multiply using a scale factor in proportional relationships and problem-solving? Explain.

6. How much apple juice will Elena need to make 40 cups of fruit punch?
15 cups

The chicken marinade recipe she uses says, "Mix 3 parts oil with 2 parts soy sauce and 1 part orange juice." Elena needs to make 13.5 cups of marinade for the amount of chicken she's marinating.

7. Create a proportion that could be used to find the amount of oil Elena needs for the recipe.

$$\frac{\text{oil}}{\text{Total}} \Bigg| \frac{3}{6} = \frac{?}{13.5 \text{ c}} \quad \boxed{6.75 \text{ cups}}$$

8. Elena is scaling the recipe up to make a larger batch. By what scale factor is Elena enlarging the recipe? Explain or show how you know.

$$\frac{13.5}{6} = 2.25 \text{ or } 2 \frac{1}{4}$$

9. How could this scale factor be used to determine how many cups of each ingredient she needs to make the marinade?

Multiply each number of parts by 2.25

Oil: $3 \times 2.25 = 6.75$ cups oil

Soy Sauce: $2 \times 2.25 = 4.5$ cups soy sauce

Orange Juice: $1 \times 2.25 = 2.25$ cups OJ

10. Complete the summary by writing numbers on each blank based on the situation.
If she's measuring in cups, Elena can make 6 cups of marinade using the original recipe. She is scaling the recipe up by a scale factor of 2.25 to make 13.5 cups of marinade. To do this, she will mix 6.75 cups of oil with 4.5 cups of soy sauce and 2.25 cups of orange juice.



6.8.3 Bring It Together: Proportions

Recall that ratios can be used to describe part-to-part relationships or part-to-whole relationships.

Situation: Elena’s fruit punch recipe uses 5 parts cranberry juice for every 3 parts apple juice.	
Part-to-Part Example	Part-to-Whole Example
The ratio of cranberry juice to apple juice is 5:3.	The ratio of cranberry juice to total fruit punch is 5:8.

When setting up proportions to solve problems, it’s important to first determine which quantities need to be compared to find the missing value.

1. Complete the table.

Situation: A teacher is planning a field trip to the aquarium. The aquarium requires 2 chaperones for every 15 students. The teacher plans accordingly and orders a total of 85 tickets.	
Set up a proportion that can be used to <u>determine the number of students</u> the teacher is planning to have on the field trip.	$\frac{15}{17} = \frac{s}{85}$
Set up a proportion that can be used to determine the <u>number of chaperones</u> the teacher is planning to have on the field trip.	$\frac{2}{17} = \frac{c}{85}$

2. Solve only one of the proportions above.

c = 10 chaperone tickets

3. Explain how the solution to the proportion can be used to determine both the number of student tickets and the number of chaperone tickets the teacher ordered.
You can use the same scale factor, 5, to find the number of student tickets.
OR
Subtract 10 from 85 to find the number of student tickets.

6.8.3 BRING IT TOGETHER: PROPORTIONS

ENRICHMENT

In your own words, describe the difference between and in this context.

ENRICHMENT

Give an example of a part: part relationship in this context. Give an example of part: whole relationship in this context.

TEACHER MOVES

First proportion: “S gives you the number of student tickets, which you can then subtract from the total of 85 to find the number of chaperone tickets.”

Second proportion: “The second proportion gives you a scale factor of 5, which you can use to multiply by the number of students (15) to find the total.”

6.8.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider extending the Three Reads language routine to these questions for students who still need support interpreting word problems and constructing proportions.

- First Read: Focus on what the question is asking.
- Second Read: Focus on units and values included.
- Third Read: Focus on representing the values and question using a proportion.



6.8.4 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar. How much oil is needed to make 19 teaspoons (tsp) of salad dressing?

$$\frac{3}{4} = \frac{x}{19} \quad \boxed{14 \frac{1}{4} \text{ tsp.}}$$

2. A grocer can buy strawberries from a local farmer for \$1.38 per pound (lb). His last strawberry order cost \$241.50. How many pounds did the grocer order?

$$\$241.50 \times \frac{1 \text{ lb.}}{\$1.38} = \boxed{175 \text{ lb.}}$$

3. Andre and Han are moving boxes. Andre can move 4 boxes every half hour. Han can move 5 boxes every half hour. How long will it take Andre and Han to move all 72 boxes?

$$\frac{30 \text{ min}}{9 \text{ boxes}} = \frac{x}{72 \text{ boxes}} \quad \boxed{\begin{array}{c} 240 \text{ minutes} \\ \text{or} \\ 4 \text{ hours} \end{array}}$$



6.8.5 Wrap-Up: Ticket Out the Door

A recipe for sparkling grape juice calls for $1\frac{1}{2}$ quarts of sparkling water and $\frac{3}{4}$ quart of grape juice.

1. How much sparkling water should be mixed with 9 quarts of grape juice?

$$\frac{1.5}{0.75} = \frac{?}{9}$$

18 quarts

2. How much grape juice should be mixed with $\frac{15}{4}$ quarts of sparkling water?

$$\frac{1.5}{0.75} = \frac{3.75}{?} \quad \frac{\frac{3}{2} \times \frac{5}{2}}{\frac{3}{4} \times \frac{5}{2}} = \frac{\frac{15}{4}}{?} = \frac{15}{8}$$

1.875 qt. or $1\frac{7}{8}$ qt.

6.8.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Students should continue to build fluency in operations with fractions. However, when evaluating student work, consider giving two scores, one for setting up proportions and one for calculations.

This differentiated approach will not only give you more specific information per student but will also give students more specific feedback on their work.